

Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCP-003
Document class	Process Characterization Plan
Title	Process Characterization Plan: Production Bioreactor (Step 3)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

2-AB, 2-aminobenzamide; ADCC, antibody-dependent cell-mediated cytotoxicity; ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; CCD, central

composite design; CHO, Chinese hamster ovary; CPP, critical process parameter; CQA, critical quality attribute; DO, dissolved oxygen; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; ELN, electronic laboratory notebook; GPP, general process parameter; HCP, host cell protein; HILIC, hydrophilic interaction liquid chromatography; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; iVCC, initial viable cell concentration; KPP, key process parameter; LIMS, laboratory information management system; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; pCO₂, partial pressure of dissolved carbon dioxide; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SDM, scale-down model; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; UPLC, ultra-performance liquid chromatography; VCD, viable cell density; WC-CPP, well-controlled critical process parameter.

1 Purpose and scope

The production bioreactor is the step at which A-Mab acquires most of the quality attributes the drug substance must meet. Fed-batch culture forms the glycan distribution, the charge variant distribution and the aggregate content of the harvested material, and the platform purification train that follows (Protein A capture, low-pH inactivation, cation exchange, anion exchange, virus filtration and formulation) modifies neither the glycan nor the charge variant distribution. Experience with the platform establishes how a humanized IgG1 culture behaves, but it does not fix the ranges over which A-Mab may be operated at the receiving site. RA-001 ranked this step as the largest single source of risk to product quality in the drug substance process, and assigned it the widest characterization scope in the campaign.

This plan defines the studies that close that gap, and it defines them before they are executed. It states which parameters will be varied and over what ranges, how the cultures will be run and sampled, which validated methods will measure each attribute, and by what criteria the operating region, the proven acceptable ranges and the parameter classifications will be decided. The studies will be executed on a qualified scale-down model of the commercial production bioreactor (§5.1), and the results, the classifications and the resulting contribution to the control strategy will be reported in PCR-003.

The plan covers the 9 process parameters listed in §6.1 and the 7 quality attributes the register assigns to this step. Four topics fall outside it. Seed expansion up to and including the N-1 bioreactor is not characterized, because RA-001 assessed it as a low risk to product quality, and the transfer of those steps is covered by the parent transfer plan (PTP-001). Harvest and clarification are covered by PCP-004, and no viral clearance is claimed for this step, so no spiking study is planned here. Confirmation at commercial scale belongs to Stage 2 and to the 5 process performance qualification batches, so nothing in this plan should be read as a claim about performance at 15,000 L.

2 Objectives

The studies described in this plan have five objectives.

1. Quantify the effect of each characterized process parameter on each quality attribute the production bioreactor forms, over the characterization ranges given in §6.1.
2. Define a multivariate operating region in the well-controlled culture parameters within which every attribute in scope is predicted to meet acceptance.
3. Establish a proven acceptable range for every parameter and attribute pair the response-surface model covers, and determine whether the normal operating range falls inside each one.
4. Classify every parameter in scope under SOP-4001, including any parameter for which no effect is found.
5. Deliver the ranges, in-process controls and monitoring requirements that the control strategy and the Stage 2 qualification will rely on.

The first three objectives are quantitative and depend on the studies succeeding, while the last two are decisions, and §7 fixes in advance the rules by which those decisions will be made.

3 Related documents

Table 3.1 lists the documents this plan depends on and the documents it feeds. PTP-001 frames the transfer of the drug substance process between sites, RA-001 sets the scope of what is characterized here, and PCMP-001 states the campaign-wide conventions that this plan instantiates for Step 3. PCR-003 will report the executed studies, and PCMR-001 will consolidate them with the rest of the train.

Table 3.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the controlled procedures and the validated analytical methods cited in this plan, and each method is matched in §5.3 to the attribute it measures.

Table 3.2: Controlled procedures and validated analytical methods for this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP

Reference	Title	Type
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The production bioreactor is a fed-batch culture of a recombinant CHO cell line in a stirred tank vessel, inoculated from the N-1 seed bioreactor and harvested after 17 days at the set-point condition. Culture pH is held by carbon dioxide sparging and base addition, temperature by the vessel jacket, and dissolved oxygen by sparged air and oxygen, and bolus nutrient feeds are added on a fixed schedule. At commercial scale the vessel is operated at 15,000 L working volume (SOP-2003). At bench scale it is operated on the qualified model described in §5.1.

Four mechanisms shape what the studies expect to find, and the ranges in §6.1 follow from them. Galactosylation depends on the activity of the Golgi galactosyltransferase and on the supply of the nucleotide sugar donor (UDP-galactose), both of which fall as a culture ages and as the extracellular environment acidifies. High mannose is set earlier in the same pathway, by how far mannosidase trimming has progressed before the antibody is secreted, which makes it sensitive to lumenal pH and to temperature. Afucosylation reflects a competition instead. Core fucosyltransferase acts on the antibody while it transits the Golgi, so a condition that slows transit raises the fucosylated fraction and lowers afucosylation. Deamidation and aggregation are not enzymatic at all, and both accumulate with the time the secreted antibody spends in an ageing broth, which is why culture duration is carried as a factor of both multivariate designs (§6) and is expected to act on several attributes at once.

Prior product knowledge supports this framing without replacing it. The platform has produced three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) in the same medium and feed system, and several of the A-Mab acceptance criteria in Table 4.1 were set from X-Mab clinical experience (CMC Biotech Working Group 2009). Since the mechanisms above are properties of the host cell line and of the culture chemistry, they may be carried across products. The effect sizes may not. They depend on the specific productivity and the glycan profile of the individual clone, so the studies characterize A-Mab directly, and prior knowledge is used only to justify the factor set and the ranges.

4.2 Quality attributes in scope

Table 4.1 lists the quality attributes the register assigns to this step, each with an acceptance criterion, a level of criticality and a Tool #1 score. Criticality is treated as a continuum and not as a binary classification (International Council for Harmonisation

2023; CMC Biotech Working Group 2009). The Tool #1 score is the product of the impact of an attribute on safety or efficacy and the uncertainty in that impact, and attributes at high or very high criticality (H or VH) are treated as critical.

Table 4.1: Quality attributes formed at the production bioreactor, with acceptance criteria, criticality and Tool #1 score.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2-13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10-40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3-10 % of glycans	VH	80
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

High mannose is the only attribute of very high criticality and carries the highest Tool #1 score in Table 4.1, so it is treated as the governing attribute of the step and the operating region will be judged against it first. Afucosylation and galactosylation come next, both linked to antibody-dependent cell-mediated cytotoxicity (ADCC), which is the primary mechanism of action of A-Mab, and both formed only in the bioreactor. Aggregate is also of high criticality. However, this step shares control of aggregate with cation exchange (PCR-007), so a harvest level inside acceptance is necessary and is not sufficient.

Acidic charge variants sit at the other end of the range, at very low criticality. The attribute is measured on every run nonetheless, since deamidation is a stability-indicating change and a large shift would be informative even where the acceptance criterion is as wide as this one.

Host cell protein and residual DNA are formed here and cleared downstream, and this step sets the load that the capture column will see, so both will be measured on the harvest of every run to characterize that load. Neither is a response of the designed experiments. The clearance credit for both belongs to Protein A capture (PCR-005), cation exchange (PCR-007) and anion exchange (PCR-008), and PCMR-001 consolidates it across the train.

4.3 Risk-based prioritization of parameters

RA-001 assessed every process parameter of this step by risk ranking and filtering. Each parameter received one score for the potential to affect a critical quality attribute and a second for the potential to interact with other parameters, and the two scores together decided the study type (CMC Biotech Working Group 2009). Parameters scoring high on both went into the multivariate design, since an interaction can only be estimated when the interacting parameters are varied together. Parameters with a credible main effect and little interaction potential were assigned to univariate assessment. No parameter of this step was left unstudied.

5 of the 9 parameters are therefore studied multivariately and 4 univariately (Table 6.1). The split is not a statement about which parameters matter, which is what the studies exist to determine. It is a statement about which parameters are expected to act through the same chemistry and so have to be varied together.

The characterization ranges are wider than the normal operating ranges throughout, by factors between 1.5 and 2.5. Two considerations set that width: a range has to be wide enough for a real effect to be resolved above assay and process noise, and wide enough to bound a knowledge space that can support later movement within the design space without new studies. It also has to stay inside what the equipment can deliver and inside conditions the culture survives. Dissolved carbon dioxide is varied over a range that is wide in proportion to the set-point (Table 6.1), because it accumulates at commercial scale in a way a bench vessel does not reproduce without deliberate control.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be executed in bench-scale stirred tank bioreactors qualified as a model of the commercial vessel (SOP-1001). Qualification is a precondition of execution and not an outcome of it, so no characterization run may start before the qualification report is approved.

The model rests on a standard division between scale-independent and scale-dependent variables: the first are set to the values the commercial process will use, and the second are set so that performance matches across scales (CMC Biotech Working Group 2009). Culture pH, temperature, dissolved oxygen, initial viable cell concentration, feed schedule and culture duration are scale-independent, and the bench vessels will run them at commercial target values. Agitation, gas flow rates, headspace pressure and working volume are scale-dependent, and each will be set to match the commercial vessel on volumetric mass transfer coefficient, impeller tip speed and the pCO₂ profile through the culture.

Qualification compares the model with at-scale data on inputs and on outputs. The input attributes are the seed train material and the medium and feed lots. The output attributes are the viable cell density profile, viability at harvest, titre and the 5 quality attributes the designed experiments will measure. The model is qualified when no statistically significant difference is found between the bench and the commercial distributions of these attributes at the set-point condition. The qualification package is filed under SOP-1001 and will be referenced, not repeated, in PCR-003.

Two limits of the model should be stated in the sections that rely on it. A bench vessel mixes in seconds where a 15,000 L vessel takes minutes, so any gradient effect at commercial scale is under-represented at bench scale. Carbon dioxide also strips more readily from a small vessel, which is why pCO₂ is controlled as an independent factor in these studies (§6.1) instead of being allowed to find a level of its own. Neither limitation is removed by this plan. Both are managed by carrying pCO₂ as a characterized factor and by confirming the attributes at commercial scale during Stage 2.

The centre point runs of both designs serve a second purpose: replicated at the set-point of every factor, they estimate pure error, which is the reference against which lack of fit is tested (§5.5).

5.2 Cell line, media and culture operation

The cell line is the A-Mab production clone, expanded from a single vial of the working cell bank through the seed train to the N-1 bioreactor. Seed expansion is run at platform conditions and is not characterized here (§1). Each characterization run will be inoculated from an N-1 culture meeting the platform criteria for viable cell density and viability, so that the input to the study is controlled and is not itself a source of variation.

Basal medium and feeds are prepared under SOP-2004 from the platform formulation. Medium and feed lots are identity tested and released against specification before use, and they are inputs controlled to specification and not variables under characterization, with one exception. Basal medium concentration is a univariate factor of this study (§6.4), and the levels will be prepared as separate lots to the same formulation.

The production culture is operated under SOP-2003, with bolus feeds added on the platform schedule and harvest at the duration the design assigns to the run. Culture duration is a factor of the multivariate designs and may not be allowed to float, so a run that reaches the platform stop criterion for viability before the assigned duration departs from the design and will be handled as a deviation (§9).

5.3 Analytical methods

Every attribute in scope is measured by a validated method, and Table 3.2 gives the procedure and the validation reference for each. Released N-glycan mapping by 2-AB HILIC-UPLC (SOP-3010, AMV-3010) reports afucosylation, galactosylation and high mannose from a single analysis, so the three glycan attributes come from one sample and share a common analytical variance. Size variants are measured by SEC-HPLC (SOP-3011, AMV-3011), which reports the high molecular weight fraction, and charge variants by icIEF (SOP-3013, AMV-3013), which reports the acidic fraction. Host cell protein is measured by ELISA (SOP-3012, AMV-3012) and residual host cell DNA by qPCR (SOP-3014, AMV-3014).

Method performance is not restated in this plan. Accuracy, precision, specificity, range and quantitation limits are given in the validation report cited for each method in Table 3.2, and PCR-003 will quote the relevant precision estimate wherever an effect is small enough for assay variance to matter.

Samples for a given attribute will be analysed in as few analytical sequences as the sample count allows, and a common reference standard will be included in every sequence. Since a design is analysed as a whole, an analytical batch effect that tracked the run order would be indistinguishable from a factor effect.

5.4 Sampling plan

Sampling has two purposes: process monitoring follows the culture while it runs, and the harvest sample characterizes what the culture produced.

Monitoring samples will be drawn daily from each vessel and tested for viable cell density, viability and the platform metabolite panel (SOP-2003). Offline pH, dissolved oxygen and pCO₂ will be measured on the same sample and compared with the online probe readings, and that comparison is the control that detects probe drift during a run.

The harvest sample is the study sample. One clarified harvest sample per run will be taken at the end of culture and tested for every attribute named in §5.3, and retained aliquots will be stored frozen so that a repeat test or an additional test can be run without repeating the culture.

Replication is concentrated at the centre point, where the screening design carries 3 runs and the response-surface design carries 4, all at the set-point of every factor. These runs estimate the reproducibility of the whole system, from culture through assay, and they supply the pure error term used in §5.5. The factorial and axial runs are not replicated, which is the usual trade in a designed experiment: precision on the effects is bought from the structure of the matrix instead of from repetition.

5.5 Statistical methods

All models will be fitted on coded factors, each coded so that the set-point maps to zero and the edges of the characterization range map to ± 1 , which puts every factor on a common scale and makes the coefficients comparable (SOP-1002).

The screening design will be analysed by ordinary least squares (OLS) with all main effects and all two-factor interactions in the model, and terms will be tested at $\alpha = 0.05$. An effect is reported as twice the coded coefficient, so it is the change in the response across the whole characterization range of the factor.

The screening model identifies the factors that matter, and it is not a predictive model: it estimates 16 terms from 19 runs, which leaves 3 residual degrees of freedom, 2 of them pure error and 1 lack of fit. A model that near-saturated fits the data it was built on closely whatever the truth is. Its coefficients will therefore be used to select factors and directions, and not to predict a response at a setting the design did not run.

The response-surface design will be analysed as a full quadratic model in the 4 factors carried forward, with all main effects, all two-factor interactions and all quadratic terms. It estimates 15 terms from 28 runs, which leaves 13 residual degrees of freedom, 3 of them pure error and 10 lack of fit. This is the predictive model, and the operating region, the proven acceptable ranges and the capability estimate are all derived from it.

Model adequacy will be reported as R^2 , adjusted R^2 , predicted R^2 from the PRESS statistic, the model F statistic and an ANOVA lack-of-fit test against pure error. Residual

diagnostics will be reported for every response: residuals against predicted values, a normal quantile plot, and predicted against actual values.

Capability at commercial scale will be estimated by simulation. 2,000 batches will be simulated by drawing each parameter from within the normal operating range, predicting each attribute from the fitted response-surface model, adding the residual variance of that model, and computing a one-sided capability index against the acceptance criterion. The simulation uses a fixed random seed, so the reported values are reproducible.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 lists every parameter in scope with the set-point, the range to be studied, the normal operating range and the study type. Table 6.2 gives the letter codes used for the multivariate factors in the design matrices and in the analysis.

Table 6.1: Process parameters of the production bioreactor, with the ranges to be studied and the study type assigned by RA-001.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
Dissolved CO2 (pCO2)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% sat	50	30–70	40–60	univariate
Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	X	1.2	0.8–1.6	1–1.4	univariate
Nutrient feed-1 volume	% WV	12	9–15	10–14	univariate

Table 6.2: Coded factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO2 (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

The two ranges in Table 6.1 do different jobs. The normal operating range is the band within which the commercial process will be controlled, and it is the band the capability simulation draws from (§5.5). The range studied is wider, and it is the edge of the knowledge space: it is where the studies look for the setting at which an attribute leaves acceptance. A proven acceptable range is determined inside the range studied and is reported per attribute (§8), so the range studied is not itself a proven acceptable range.

Culture pH, culture temperature, dissolved carbon dioxide, osmolality and culture duration are the multivariate factors. All five act on the same intracellular chemistry, either through the pH and temperature dependence of the Golgi glycosyltransferases or through the time the antibody spends in culture, so interactions between them are expected and have to be estimable. By contrast, dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed volume act mainly on cell growth and nutrient supply. RA-001 scored the potential for these four to interact with the glycosylation chemistry as low, and they are assessed one parameter at a time (§6.4).

6.2 Screening design

The screening study is a two-level fractional factorial design in the 5 multivariate factors augmented with centre points, comprising 16 factorial runs and 3 centre point runs, or 19 runs in all. The fraction is a half replicate of Resolution V, so every main effect and every two-factor interaction is estimable clear of the others, and only interactions of three factors and above are aliased with the terms of interest (Appendix A).

Runs will be executed in randomized order across the available vessels, with the centre point runs distributed through the campaign instead of grouped. Randomization is the control against a time trend in the vessels, in the medium lots or in the assay.

The purpose of the screening study is to identify which factors and which interactions have an effect large enough to matter, and in which direction. It is not intended to produce a model that predicts (§5.5).

6.3 Response-surface design

The response-surface study is a face-centred central composite design (CCD) in 4 of the screening factors, comprising 16 factorial runs, 8 axial runs and 4 centre point runs, or 28 runs in all. The axial points sit on the faces of the coded cube instead of outside it, so no run is executed beyond the characterization ranges in Table 6.1, and that choice gives up some rotatability to buy a design that stays inside conditions the vessels can hold and the culture can survive (Appendix B).

The factors carried forward are culture pH, culture temperature, culture duration and dissolved carbon dioxide. Although the matrix in Appendix B is written on the

assumption that screening confirms these four, screening may identify a different set. The matrix will then be re-issued before execution, which is an amendment to this plan and not a deviation from it.

This design adds the curvature terms the screening design cannot estimate. It is the source of the operating region in §7, of the proven acceptable ranges in §8 and of the capability estimate in §5.5.

6.4 Univariate assessment

Dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed volume will be assessed one parameter at a time. Each will be run at the low edge and at the high edge of the characterization range in Table 6.1, with every other parameter held at the set-point, which gives 8 culture conditions in all. The centre point runs of the screening design provide the set-point comparator, since they already sit at the set-point of every parameter in the study.

A study that varies one parameter at a time cannot detect an interaction between the parameter varied and the parameters held fixed. That limitation is accepted on the risk basis in §4.3, where these four parameters were scored as unlikely to interact with the chemistry that forms the glycan attributes. However, if a univariate result contradicts that expectation, the parameter will be added to a follow-up multivariate design, and the addition will be reported in PCR-003.

A parameter with no detectable effect is still a result. Any such parameter will be classified under SOP-4001 on the evidence of the range it was held over, and that range will be retained in the knowledge space as evidence that the attributes are robust to the parameter.

7 Acceptance and decision criteria

The studies will be judged against the rules in this section, and the rules are stated before execution so that the judgement does not depend on what the data turn out to be.

The scale-down model is qualified when the bench and the commercial distributions of the input and output attributes listed in §5.1 show no statistically significant difference at $\alpha = 0.05$ at the set-point condition. Where a difference is found, the model will be modified and requalified before any characterization run is executed. A model that cannot be qualified for one attribute may still be used for the others, provided PCR-003 states which claims it does not support.

A fitted model is adequate when the lack-of-fit test against pure error is not significant at $\alpha = 0.05$, and when the adjusted and predicted R^2 are reported beside R^2 so that over-fitting is visible. Predicted R^2 is the most informative of the three here, since it is computed by leaving each run out of the fit in turn. A model that fails the lack-of-fit test will not be used to define an operating region, and the response concerned will instead be augmented with further runs or reported as characterized univariately, with PCR-003 stating which was done and why.

A term is retained in a model when it is significant at $\alpha = 0.05$, and a term that is not significant may still be retained where it is the lower order term of a significant interaction or quadratic, because dropping it changes the interpretation of the term above it. Terms retained on that basis will be identified as such.

The operating region will be declared over the part of the characterized space in which the response-surface model predicts that every attribute the step governs meets acceptance. It will be reported as a multivariate region and not as a set of independent parameter ranges, since the region is bounded by interactions between the factors. Movement within the region is not a change to the process, and movement outside it requires change control (International Council for Harmonisation 2009, 2008).

Capability will be reported as a one-sided index for every attribute in scope, with the margin between the simulated mean and the acceptance limit stated alongside it. No numeric minimum for the index is fixed by this plan. That criterion is defined per study, because a meaningful minimum depends on the criticality of the attribute and on how detectable an excursion would be, and neither of those is a property of the simulation.

Each parameter will then be classified under SOP-4001 into one of four classes. A parameter is a critical process parameter (CPP) when variation within the ranges studied affects a critical quality attribute and there is a real risk of operating outside the design space. It is a well-controlled critical process parameter (WC-CPP) when the same impact exists but the parameter is easy to hold, so the risk of an excursion is

low. It is a key process parameter (KPP) when it affects process performance without affecting product quality. It is a general process parameter (GPP) when no impact on a critical quality attribute is demonstrated over the range studied. Classification rests on the demonstrated effect and on the control capability together, and PCR-003 will give the reason for each parameter.

The study as a whole passes when the scale-down model is qualified, when an adequate model exists for every attribute the step governs, when an operating region containing the whole normal operating range can be declared, and when every parameter carries a classification with a stated reason. Failure against any of these is not a failure of the plan. It will be reported in PCR-003 as a finding, together with the action taken.

8 Proven acceptable ranges (planned analysis)

The operating region defined in §7 is multivariate, and a multivariate region is an awkward thing to write into a batch record. Each parameter will therefore also be reported with a proven acceptable range (PAR) for every attribute the response-surface model shows it to govern. A proven acceptable range is the univariate form of the same information.

The acceptance basis comes first: each attribute will be judged against the drug substance acceptance criterion in Table 4.1. Where the control strategy establishes an in-process limit on the harvest for an attribute, the proven acceptable range will be determined against that limit instead, because an intermediate has to meet a criterion that applies to the intermediate. PCR-003 will state which criterion was used for each attribute.

Two analyses will be run for every pair of a parameter and an attribute. The first holds the other factors at their set-points and scans the parameter of interest across the full characterization range, evaluating the fitted response-surface model on a grid of 81 points. The second propagates the normal operating ranges of the other factors: at each grid point, 2,000 values of every remaining factor are drawn from within the normal operating range, the fitted model is evaluated at each draw, the residual variance of the model is added, and the 95 % predictive interval of the attribute is computed.

The decision rule differs between the two analyses, and it does so deliberately. In the first, the proven acceptable range is the contiguous span of the parameter, containing the set-point, over which the predicted mean stays within acceptance. In the second it is the contiguous span over which the whole 95 % predictive interval stays within acceptance. The second is a robustness criterion and will normally give the narrower range. Where the two coincide across the entire characterization range, the parameter is acceptable everywhere it was studied and is robust to the rest of the operating point moving within the normal operating range as well.

Every governed attribute will be reported with a two-panel plot against the governing parameter, which is the response-surface factor carrying the largest main effect on that attribute. The panels show the two analyses side by side, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked.

The relationship to the operating region should be read carefully. A proven acceptable range is univariate, so it holds for the one parameter it names under the conditions of the analysis that produced it. It is not a licence to move two parameters to opposite

edges at once. Where a proven acceptable range covers the whole characterization range, the binding constraint on the process is the corner of the multivariate region and not that range. Both results are bounded by the characterization ranges in Table 6.1, by the fitted response-surface model and by the conditions the scale-down model represents (§5.1).

9 Data management and integrity

Data generated under this plan are held to the same integrity requirements as manufacturing records. Raw data are attributable, legible, contemporaneous, original and accurate, and they are complete, consistent, enduring and available.

Culture data are captured by the bioreactor control system and retained as the raw record. Analytical data are captured in the laboratory information management system (LIMS), and the calculations that turn instrument output into a reportable result are performed by validated templates. Study design, run assignment and statistical analysis are recorded in the electronic laboratory notebook (ELN), and the analysis scripts are held under version control, so a reported number can be traced back to the data and to the code that produced it.

Every record is reviewed by a second qualified person before it is used, and a correction leaves the original entry legible and carries the reason and the date. Audit trails are enabled on all systems and are reviewed as part of that data review.

This plan is a controlled document, so a change to the design, to the ranges or to the acceptance criteria after approval requires an approved amendment before the affected runs are executed. A departure discovered after execution is recorded as a deviation, investigated for root cause, and assessed for impact on the study conclusions in PCR-003.

10 Roles and responsibilities

Table 10.1 assigns the work across the functions that execute and approve this plan. Process Development owns the plan, the execution and the report, and Manufacturing Science and Technology (MSAT) owns the scale-down model and the link to the receiving site. Analytical Development owns the methods and the analytical data, Statistics owns the designs and the model fitting, and Quality Assurance approves both this plan and PCR-003.

Table 10.1: Functions and responsibilities for the execution of this plan.

Function	Responsibility
Process Development	Authors this plan, executes the cultures, interprets the results, writes PCR-003
Manufacturing Science and Technology	Qualifies and maintains the scale-down model; confirms that set-points and ranges are implementable at the receiving site
Analytical Development	Performs the assays in §5.3, maintains method validation, reports analytical performance
Statistics	Approves the designs, performs the model fitting, the capability simulation and the proven-acceptable-range analysis
Quality Assurance	Approves this plan and PCR-003, dispositions deviations, confirms data integrity
Manufacturing, receiving site	Reviews the resulting ranges and control requirements for commercial implementability

11 Deliverables and schedule

The deliverable of this plan is PCR-003, which will report the executed studies, the operating region, the proven acceptable ranges, the capability estimate and the parameter classifications. The scale-down model qualification report is a prerequisite deliverable and is filed separately under SOP-1001.

The work runs in a fixed order, with model qualification first because it gates everything after it. The screening study follows, with 19 runs, and analysis of the screening data then confirms or revises the factors carried into the response-surface design, with any revision issued as an amendment before execution (§6.3). The response-surface study follows, with 28 runs. Once the screening result is available the univariate conditions of §6.4 may be executed in parallel with it, since they do not depend on the response-surface runs. Analysis, capability simulation, classification and reporting come last.

55 cultures are planned in all. No calendar dates are fixed here: the sequence is fixed, and the campaign schedule is maintained under PCMP-001.

12 Risks and assumptions

Three assumptions carry the study, and each has a risk attached to it.

The first is that the scale-down model represents the commercial vessel closely enough for a conclusion drawn at bench scale to hold at 15,000 L. Qualification tests that at the set-point condition only, and it does not test it at the edges of the characterization ranges, so the assumption is weakest exactly where the operating region is most constrained. The risk is managed by carrying pCO₂ as a characterized factor, since it is the parameter that differs most between the two scales, and by confirming the attributes at commercial scale during Stage 2.

The second is that screening identifies the right subset of factors to carry into the response-surface design. A factor with a real effect that screening failed to resolve would be excluded from the predictive model and would not appear in the operating region at all. The risk is managed in two ways: every factor not carried forward is held at the set-point through the response-surface runs, so an excluded factor contributes no variation to the fitted model, and each is classified on the screening evidence with the range that evidence supports.

The third is that platform and prior product knowledge transfer to A-Mab. The mechanisms transfer, because they are properties of the host cell line and of the culture chemistry, but the effect sizes may not, because they depend on the clone. As a result the studies characterize A-Mab directly, and prior knowledge is used only for the factor set and the ranges (§4.1).

Two operational risks remain. A change of medium or feed lot during a campaign of 55 cultures could introduce a step change confounded with run order, and the controls are randomization (§6.2) and the use of a single lot per component wherever the campaign length allows. Assay variability may also be large in comparison with a small effect near an acceptance limit, and the controls there are the replicated centre points, which estimate the combined culture and assay variance directly, and the precision estimates in the method validation reports.

13 Approvals

This plan takes effect when it is approved by the functions listed in Table 1. Approval signifies that the study design, the ranges, the analytical methods and the decision criteria are agreed before execution, and that a change to any of them requires an amendment under §9.

Appendix A — Planned screening design matrix

The matrix below gives the planned screening runs in coded levels, using the letter codes of Table 6.2. Response columns are absent because no run has been executed.

Table 13.1: Planned screening design matrix, coded levels.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

Appendix B — Planned response-surface design matrix

The matrix below gives the planned response-surface runs in coded levels for the factors carried forward under §6.3. It will be re-issued if screening identifies a different set.

Table 13.2: Planned response-surface design matrix, coded levels.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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